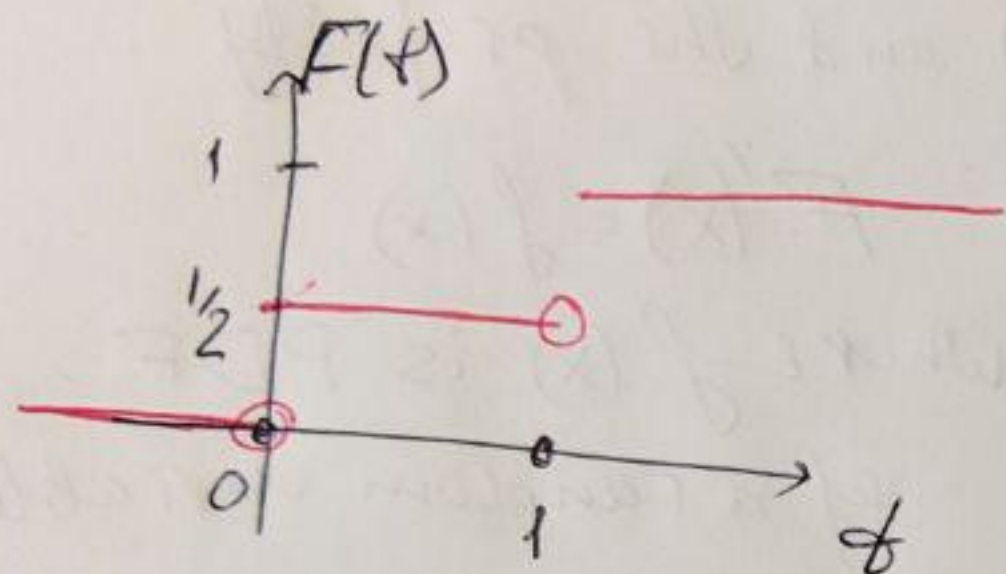


Then $P(X \leq t) = 0$ if $t < 0$

$P(X \leq t) = \frac{1}{2}$ if t is between 0 and 1
(only the case $X=0$ works)

$P(X \leq t) = 1$ if $t \geq 1$ (always true ~~if~~ since X is always ≤ 1)



Answer:

$$F(t) = \begin{cases} 0 & \text{if } t < 0 \\ \frac{1}{2} & \text{if } t \in [0, 1) \\ 1 & \text{if } t \in [1, +\infty) \end{cases}$$

Example 2

CDF of exponential random variable

$$X \sim \text{Exp}(\lambda)$$

$$F(t) = \int_{-\infty}^t f(x) dx, \quad f(x) = \lambda e^{-\lambda x} \quad \text{if } x > 0 \quad \text{-exponential PDF}$$

$$\bullet \text{ if } t > 0 \quad F(t) = \int_0^t \lambda e^{-\lambda x} dx = e^{-\lambda x} \Big|_0^t = 1 - e^{-\lambda t}$$

$\bullet$ if $t \leq 0$ $P(X \leq t) = 0$ since X is always ≥ 0 .

Answer:

$$F(t) = \begin{cases} 1 - e^{-\lambda t} & \text{if } t > 0 \\ 0 & \text{otherwise} \end{cases}$$